Gas.	Boiling-point.	Solubility in water.
Hydrogen	−253° C.	19 parts in 1000.
Carbon monoxide	−190	25
Oxygen	−183	34
Methane	−164	--
Carbon dioxide..............	− 65	1019
Ammonia	− 33·5	802,000
Sulphur dioxide	− 10	56,600

Summary.

In the foregoing paragraphs an attempt has been made to show the shortcomings of an unamplified columnar theory in explaining all the phenomena of "Initial Recombination." The arguments are based on the failure of mathematical deductions of the theory, whether tested directly or indirectly, to predict experimental results; and also on the great diversity shown by various gases with respect to this phenomenon. Thus ammonia, a light gas, should theoretically show little recombination, but it is found to show "initial recombination" in a very marked degree. Moisture was found to enhance recombination, which is attributed to the decrease in mobility of the ions due to the presence of moisture. The suggested reason for the deviation from the exact laws is the presence of a small percentage of ions of small mobility in the gases. This receives support from the fact that the nearer the temperature of a gas is to its boiling-point, the more does the gas exhibit initial recombination.

My thanks are due to Professor Bragg and Dr. Campbell for invaluable assistance and encouragement during the time in which the foregoing work was in progress.

XC. *On the Stability of the Laminar Motion of an Inviscid Fluid.* By Lord RAYLEIGH, *O.M., F.R.S.**

THE equations of motion of an inviscid fluid are satisfied by a motion such that U, the velocity parallel to x, is an arbitrary function of y only, while the other component velocities V and W vanish. The motion may be supposed to be limited by two fixed plane walls for each of which y has a constant value. In order to investigate the stability

* Communicated by the Author.

of the motion, we superpose upon it a two-dimensional disturbance u, v, where u and v are regarded as small. If the fluid is incompressible,

$$\frac{du}{dx}+\frac{dv}{dy}=0\,; \quad . \quad . \quad . \quad . \quad . \quad . \quad . \qquad (1)$$

and if the squares and products of small quantities are neglected, the hydrodynamical equations give *

$$\left(\frac{d}{dt}+\mathrm{U}\frac{d}{dx}\right)\left(\frac{du}{dy}-\frac{dv}{dx}\right)+\frac{d^2\mathrm{U}}{dy^2}v=0. \quad . \quad . \qquad (2)$$

From (1) and (2), if we assume that as functions of t and x, u, v are proportional to $e^{i(nt+kx)}$, where k is real and n may be real or complex,

$$\left(\frac{n}{k}+\mathrm{U}\right)\left(\frac{d^2v}{dy^2}-k^2v\right)-\frac{d^2\mathrm{U}}{dy^2}v=0. \quad . \quad . \quad . \qquad (3)$$

In the paper quoted it was shown that under certain conditions n could not be complex; and it may be convenient to repeat the argument. Let

$$n/k=p+iq, \qquad v=\alpha+i\beta,$$

where p, q, α, β are real. Substituting in (3) and equating separately to zero the real and imaginary parts, we get

$$\frac{d^2\alpha}{dy^2}=k^2\alpha+\frac{d^2\mathrm{U}}{dy^2}\,\frac{(p+\mathrm{U})\alpha+q\beta}{(p+\mathrm{U})^2+q^2},$$

$$\frac{d^2\beta}{dy^2}=k^2\beta+\frac{d^2\mathrm{U}}{dy^2}\,\frac{-q\alpha+(p+\mathrm{U})\beta}{(p+\mathrm{U})^2+q^2}\,;$$

whence if we multiply the first by β and the second by α and subtract,

$$\frac{d}{dy}\left(\beta\frac{d\alpha}{dy}-\alpha\frac{d\beta}{dy}\right)=\frac{d^2\mathrm{U}}{dy^2}\,\frac{q(\alpha^2+\beta^2)}{(p+\mathrm{U})^2+q^2}. \quad . \quad . \qquad (4)$$

At the limits, corresponding to finite or infinite values of y, we suppose that v, and therefore both α and β, vanish. Hence when (4) is integrated with respect to y between these limits, the left-hand member vanishes and we infer that q also must vanish unless $d^2\mathrm{U}/dy^2$ changes sign. Thus in the motion between walls if the velocity curve, in which U is ordinate and y abscissa, be of one curvature throughout, n must be wholly real; otherwise, so far as this argument

* Proceedings of London Mathematical Society, vol. xi. p. 57 (1880); Scientific Papers, i. p. 485. Also Lamb's 'Hydrodynamics,' § 345.

shows, n may be complex and the disturbance exponentially unstable.

Two special cases at once suggest themselves. If the motion be that which is possible to a viscous fluid moving steadily between two fixed walls under external pressure or impressed force, so that for example $U=y^2-b^2$, d^2U/dy^2 is a finite constant, and complex values of n are clearly excluded. In the case of a simple shearing motion, exemplified by $U=y$, $d^2U/dy^2=0$, and no inference can be drawn from (4). But referring back to (3), we see that in this case if n be complex,

$$\frac{d^2v}{dy^2}-k^2v=0 \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (5)$$

would have to be satisfied over the whole range between the limits where $v=0$. Since such satisfaction is not possible, we infer that here too a complex n is excluded.

It may appear at first sight as if real, as well as complex, values of n were excluded by this argument. But if n be such that $n/k+U$ vanishes anywhere within the range, (5) need not there be satisfied. In other words, the arbitrary constants which enter into the solution of (5) may there change values, subject only to the condition of making v continuous. The terminal conditions can then be satisfied. Thus any value of $-n/k$ is admissible which coincides with a value of U to be found within the range. But other real values of n are excluded.

Let us now examine how far the above argument applies to real values of n, when d^2U/dy^2 in (3) does not vanish throughout. It is easy to recognize that here also any value of $-kU$ is admissible, and for the same reason as before, viz., that when $n+kU=0$, dv/dy may be discontinuous. Suppose, for example, that there is but one place where $n+kU=0$. We may start from either wall with $v=0$ and with an arbitrary value of dv/dy and gradually build up the solutions inwards so as to satisfy (3)*. The process is to be continued on both sides until we come to the place where $n+kU=0$. The two values there found for v and for dv/dy will presumably disagree. But by suitable choice of the relative initial values of dv/dy, v may be made continuous, and (as has been said) a discontinuity in dv/dy does not interfere with the satisfaction of (3). If there are other places where U has the same

* Graphically, the equation directs us with what curvature to proceed at any point already reached.

value, dv/dy may there be either continuous or discontinuous. Even when there is but one place where $n+k\mathrm{U}=0$ with the proposed value of n, it may happen that dv/dy is there continuous.

The argument above employed is not interfered with even though U is such that $d\mathrm{U}/dy$ is here and there discontinuous, so as to make $d^2\mathrm{U}/dy^2$ infinite. At any such place the necessary condition is obtained by integrating (3) across the discontinuity. As was shown in my former paper (*loc. cit.*), it is

$$\left(\frac{n}{k}+\mathrm{U}\right).\Delta\left(\frac{dv}{dy}\right)-\Delta\left(\frac{d\mathrm{U}}{dy}\right).v=0, \quad . \quad . \quad . \quad (6)$$

Δ being the symbol of finite differences; and by (6) the corresponding sudden change in dv/dy is determined.

It appears then that any value of $-k\mathrm{U}$ is a possible value of n. Are other real values admissible? If so, $n+k\mathrm{U}$ is of one sign throughout. It is easy to see that if $d^2\mathrm{U}/dy^2$ has throughout the same sign as $n+k\mathrm{U}$, no solution is possible. I propose to prove that no solution is possible in any case if $n+k\mathrm{U}$, being real, is of one sign throughout.

If U′ be written for $\mathrm{U}+n/k$, our equation (3) takes the form

$$\mathrm{U}'\frac{d^2v}{dy^2}-v\frac{d^2\mathrm{U}'}{dy^2}=k^2\mathrm{U}'v, \quad . \quad . \quad . \quad . \quad (7)$$

or on integration with respect to y,

$$\mathrm{U}'\frac{dv}{dy}-v\frac{d\mathrm{U}'}{dy}=\mathrm{K}+k^2\int_0^y\mathrm{U}'v\,dy, \quad . \quad . \quad . \quad (8)$$

where K is an arbitrary constant. Assume $v=\mathrm{U}'v'$; then

$$\frac{dv'}{dy}=\frac{\mathrm{K}}{\mathrm{U}'^2}+\frac{k^2}{\mathrm{U}'^2}\int_0^y v'\mathrm{U}'^2\,dy\,; \quad . \quad . \quad . \quad . \quad (9)$$

whence, on integration and replacement of v,

$$v=\mathrm{H}\mathrm{U}'+\mathrm{K}\mathrm{U}'\int_0^y\frac{dy}{\mathrm{U}'^2}+k^2\mathrm{U}'\int_0^y\frac{dy}{\mathrm{U}'^2}\int_0^y\mathrm{U}'v\,dy, \quad . \quad (10)$$

H denoting a second arbitrary constant.

In (10) we may suppose y measured from the first wall, where $v=0$. Hence, unless U′ vanish with y, $\mathrm{H}=0$. Also from (8) when $y=0$,

$$\left(\mathrm{U}'\frac{dv}{dy}\right)_0=\mathrm{K}. \quad . \quad . \quad . \quad . \quad . \quad . \quad (11)$$

Let us now trace the course of v as a function of y, starting from the wall where $y=0$, $v=0$; and let us suppose first that U' is everywhere positive. By (11) K has the same sign as $(dv/dy)_0$, that is the same sign as the early values of v. Whether this sign be positive or negative, v as determined by (10) cannot again come to zero. If, for example, the initial values of v are positive, both (remaining) terms in (10) necessarily continue positive; while if v begins by being negative, it must remain finitely negative. Similarly, if U' be everywhere negative, so that K has the opposite sign to that of the early values of v, it follows that v cannot again come to zero. No solution can be found unless U' somewhere vanishes, that is unless n coincides with some value of $-k\mathrm{U}$.

In the above argument U', and therefore also n, is supposed to be *real*, but the formula (10) itself applies whether n be real or complex. It is of special value when k is very small, that is when the wave-length along x of the disturbance is very great; for it then gives v explicitly in the form

$$v = \mathrm{K}(\mathrm{U}+n/k)\int_0^y \frac{dy}{(\mathrm{U}+n/k)^2}. \quad . \quad . \quad . \quad (12)$$

When k is small, but not so small as to justify (12), a second approximation might be found by substituting from (12) in the last term of (10).

If we suppose in (12) that the second wall is situated at $y=l$, n is determined by

$$\int_0^l \frac{dy}{(\mathrm{U}+n/k)^2} = 0. \quad . \quad . \quad . \quad . \quad . \quad (13)$$

The integrals (12), (13) must not be taken through a place where $\mathrm{U}+n/k=0$, as appears from (8). We have already seen that any value of n for which this can occur is admissible. But (13) shows that no other real value of n is admissible; and it serves to determine any complex values of n.

In (13) suppose (as before) that $n/k = p+iq$; then separating the real and imaginary parts, we get

$$\int_0^l \frac{(p+\mathrm{U})^2-q^2}{\{(p+\mathrm{U})^2+q^2\}^2}\,dy = 0, \quad \int_0^l \frac{q(p+\mathrm{U})\,dy}{\{(p+\mathrm{U})^2+q^2\}^2} = 0, \quad (14)$$

from the second of which we may infer that if q be finite, $p+\mathrm{U}$ must change sign, as we have already seen that it must do when $q=0$. In every case then, when k is small, the *real part* of n must equal some value of $-k\mathrm{U}$ *.

It may be of interest to show the application of (13) to a case formerly treated† in which the velocity-curve is made up of straight portions and is anti-symmetrical with respect to the point lying midway between the two walls, now taken as origin of y. Thus on the positive side

$$\text{from} \quad y=0 \quad \text{to } y=\tfrac{1}{2}b', \qquad \mathrm{U}=\frac{\mathrm{V}y}{\frac{1}{2}b'};$$

$$\text{from} \quad y=\tfrac{1}{2}b' \text{ to } y=\tfrac{1}{2}b'+b, \quad \mathrm{U}=\frac{\mathrm{V}y}{\frac{1}{2}b'}+\mu\mathrm{V}(y-\tfrac{1}{2}b');$$

while on the negative side U takes symmetrically the opposite values. Then if we write $n/k\mathrm{V}=n'$, (13) becomes

$$0=\int_0^{\frac{1}{2}b'}\frac{dy}{(2y/b+n')^2}+\int_{\frac{1}{2}b'}^{\frac{1}{2}b'+b}\frac{dy}{\{2y/b+\mu(y-\frac{1}{2}b')+n'\}^2}$$

$+$same with n' reversed.

Effecting the integrations, we find after reduction

$$n'^2=\frac{n^2}{k^2\mathrm{V}^2}=\frac{2b+b'+2\mu b(b+b')+\mu^2b^2b'}{2b+b'}, \qquad (15)$$

in agreement with equation (23) of the paper referred to when k is there made small. Here n, if imaginary at all, is a pure imaginary, and it is imaginary only when μ lies between $-1/b$ and $-1/b-2/b'$. The regular motion is then exponentially unstable.

In the only unstable cases hitherto investigated the velocity-curve is made up of straight portions meeting at finite angles, and it may perhaps be thought that the instability has its origin in this discontinuity. The method now under discussion disposes of any doubt. For obviously in (13) it can make no important difference whether $d\mathrm{U}/dy$ is discontinuous or not. If a motion is definitely unstable in the former case, it cannot become stable merely by easing off the finite angles in the velocity-curve. There exist, therefore, exponentially unstable motions in which both U and

* By the method of a former paper "On the question of the Stability of the Flow of Fluids" (Phil. Mag. vol. xxxiv. p. 59, 1892; Scientific Papers, iii. p. 579) the conclusion that $p+\mathrm{U}$ must change sign may be extended to the problem of the simple shearing motion between two parallel walls of a *viscous* fluid, and this whatever may be the value of k.

† Proc. Lond. Math. Soc. vol. xix. p. 67 (1887); Scientific Papers, iii. p. 20, figs. (3), (4), (5).

dU/dy are continuous. And it is further evident that any proposed velocity-curve may be replaced approximately by straight lines as in my former papers.

The fact that n in equation (15) appears only as n^2 is a simple consequence of the anti-symmetrical character of U. For if in (13) we measure y from the centre and integrate between the limits $\pm\frac{1}{2}l$, we obtain in that case

$$\int_0^{\frac{1}{2}l} \frac{n^2/k^2+U^2}{(n^2/k^2-U^2)^2}dy=0, \quad . \quad . \quad . \quad . \quad (16)$$

in which only n^2 occurs. But it does not appear than n^2 is necessarily real, as happens in (15).

Apart from such examples as were treated in my former papers in which d^2U/dy^2 vanishes except at certain definite places, there are very few cases in which (3) can be solved analytically. If we suppose that $v=\sin(\pi y/l)$, vanishing when $y=0$ and when $y=l$, and seek what is then admissible for U, we get

$$U+n/k=A\cos\{k^2+\pi^2/l^2\}^{\frac{1}{2}}y+B\sin\{k^2+\pi^2/l^2\}y, \quad (17)$$

in which A and B are arbitrary and n may as well be supposed to be zero. But since U varies with k, the solution is of no great interest.

In estimating the significance of our results respecting stability, we must of course remember that the disturbance has been assumed to be and to remain infinitely small. Where stability is indicated, the magnitude of the admissible disturbance may be very restricted. It was on these lines that Kelvin proposed to explain the apparent contradiction between theoretical results for an inviscid fluid and observation of what happens in the motion of real fluids which are all more or less viscous. Prof. M'F. Orr has carried this explanation further*. Taking the case of a simple shearing motion between two walls, he investigates a composite disturbance, periodic with respect to x but not with respect to t, given initially as

$$v=B\cos lx\cos my, \quad . \quad . \quad . \quad . \quad . \quad (18)$$

and he finds, equation (38), that when m is large the disturbance may increase very much, though ultimately it comes to zero. Stability in the mathematical sense (B infinitely small) may thus be not inconsistent with a practical

* Proc. Roy. Irish Academy, vol. xxvii. Section A, No. 2, 1907. Other related questions are also treated.

instability. A complete theoretical proof of instability requires not only a method capable of dealing with finite disturbances but also a definition, not easily given, of what is meant by the term. In the case of stability we are rather better situated, since by absolute stability we may understand complete recovery from disturbances of any kind however large, such as Reynolds showed to occur in the present case when viscosity is paramount *. In the absence of dissipation, stability in this sense is not to be expected.

Another manner of regarding the present problem of the shearing motion of an inviscid fluid is instructive. In the original motion the vorticity is constant throughout the whole space between the walls. The disturbance is represented by a superposed vorticity, which may be either positive or negative, and this vorticity everywhere *moves with the fluid*. At any subsequent time the same vorticities exist as initially; the only question is as to their distribution. And when this distribution is known, the whole motion is determined. Now it would seem that the added vorticities will produce most effect if the positive parts are brought together, and also the negative parts, as much as is consistent with the prescribed periodicity along x, and that even if this can be done the effect cannot be out of proportion to the magnitude of the additional vorticities. If this view be accepted, the temporary large increase in Prof. Orr's example would be attributed to a specially unfavourable distribution initially in which (m large) the positive and negative parts of the added vorticities are closely intermingled. We may even go further and regard the subsequent tendency to evanescence, rather than the temporary increase, as the normal phenomenon. The difficulty in reconciling the observed behaviour of actual fluids with the theory of an inviscid fluid still seems to me to be considerable, unless indeed we can admit a distinction between a fluid of infinitely small viscosity and one of none at all.

At one time I thought that the instability suggested by observation might attach to the stages through which a viscous liquid must pass in order to acquire a uniform shearing motion rather than to the final state itself. Thus in order to find an explanation of "skin friction" we may suppose the fluid to be initially at rest between two infinite fixed walls, one of which is then suddenly made to move in its own plane with a uniform velocity. In the earlier stages the other wall has no effect and the problem is one considered by Fourier in connexion with the conduction of heat. The

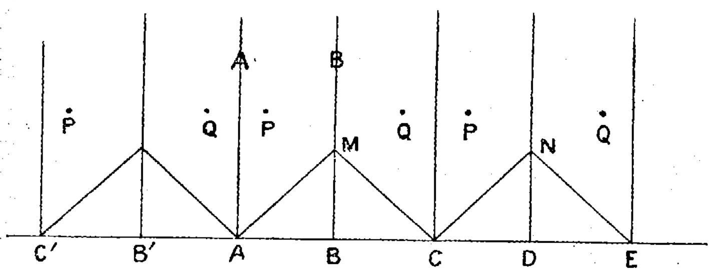
* See also Orr, Proc. Roy. Irish Acad. 1907, p. 124.

velocity U in the laminar motion satisfies generally an equation of the form

$$\frac{dU}{dt}=\frac{d^2U}{dy^2}, \qquad (19)$$

with the conditions that initially ($t=0$) $U=0$, and that from $t=0$ onwards $U=1$ when $y=0$, and (if we please) $U=0$ when $y=l$. We might employ Fourier's solution, but all that we require follows at once from the differential equation itself. It is evident that dU/dt, and therefore d^2U/dy^2, is everywhere positive and accordingly that a non-viscous liquid, moving laminarly as the viscous fluid moves in any of these stages, is stable. It would appear then that no explanation is to be found in this direction.

Hitherto we have supposed that the disturbance is periodic as regards x, but a simple example, not coming under this head, may be worthy of notice. It is that of the disturbance due to a single vortex filament in which the vorticity differs from the otherwise uniform vorticity of the neighbouring fluid. In the figure the lines AA, BB represent the situation



of the walls and AM the velocity-curve of the original shearing motion rising from zero at A to a finite value at M. For the present purpose, however, we suppose material walls to be absent, but that the same effect (of prohibiting normal motion) is arrived at by suitable suppositions as to the fluid lying outside and now imagined infinite. It is only necessary to continue the velocity-curve in the manner shown AMCN..., the vorticity in the alternate layers of equal width being equal and opposite. Symmetry then shows that under the operation of these vorticities the fluid moves as if AA, BB, &c. were material walls.

We have now to trace the effect of an additional vorticity, supposed positive, at a point P. If the wall AA were alone concerned, its effect would be imitated by the introduction of an opposite vorticity at the point Q which is the image of P in AA. Thus P would move under the influence of the original vorticities, already allowed for, and of the negative vorticity at Q. Under the latter influence it would move parallel to AA with a certain velocity, and for the same reason Q would move similarly, so that PQ would remain perpendicular to AA. To take account of both walls the more complicated arrangement shown in the figure is necessary, in which the points P represent equal positive vorticities and Q equal negative vorticities. The conditions at both walls are thus satisfied ; and as before all the vortices P, Q move under each other's influence so as to remain upon a line perpendicular to AA. Thus, to go back to the original form of the problem, P moves parallel to the walls with a constant velocity, and no change ensues in the character of the motion—a conclusion which will appear the more remarkable when we remember that there is no limitation upon the magnitude of the added vorticity.

The same method is applicable—in imagination at any rate—whatever be the distribution of vorticities between the walls, and the corresponding velocity at any point is determined by quadratures on Helmholtz's principle. The new positions of all the vorticities after a short time is thus found, and then a new departure may be taken, and so on indefinitely.

XCI. *An Electromagnetic Hypothesis as to the Origin of Series Spectra.* By ARTHUR W. CONWAY, *M.A., D.Sc., Professor of Mathematical Physics, University College, Dublin* *.

INTRODUCTION.

THE following paper is an attempt to picture a model of an atom based on the classical electrodynamics which will illustrate some of the properties of spectral series. The principal difficulties in explaining a frequency formula of the type

$$\nu = A - B/(n+\mu)^2,$$

where

$$n = 1, 2, 3 \ldots\ldots,$$

are three: (i.) The formula gives the frequency and not the square of the frequency ; (ii.) no elastic or electrical system has a frequency equation of this type, the frequency

* Communicated by the Author.